INCH-POUND

MIL-C-55302/139B(USAF)
11 Feb 93
SUPERSEDING
MIL-C-55302/139A(USAF)
27 March 1986

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE, SOCKET CONTACTS, 160 CONTACT POSITIONS FOR PRINTED WIRING BOARDS (100 INCH SPACING)

This specification is approved for use by the Rome air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302

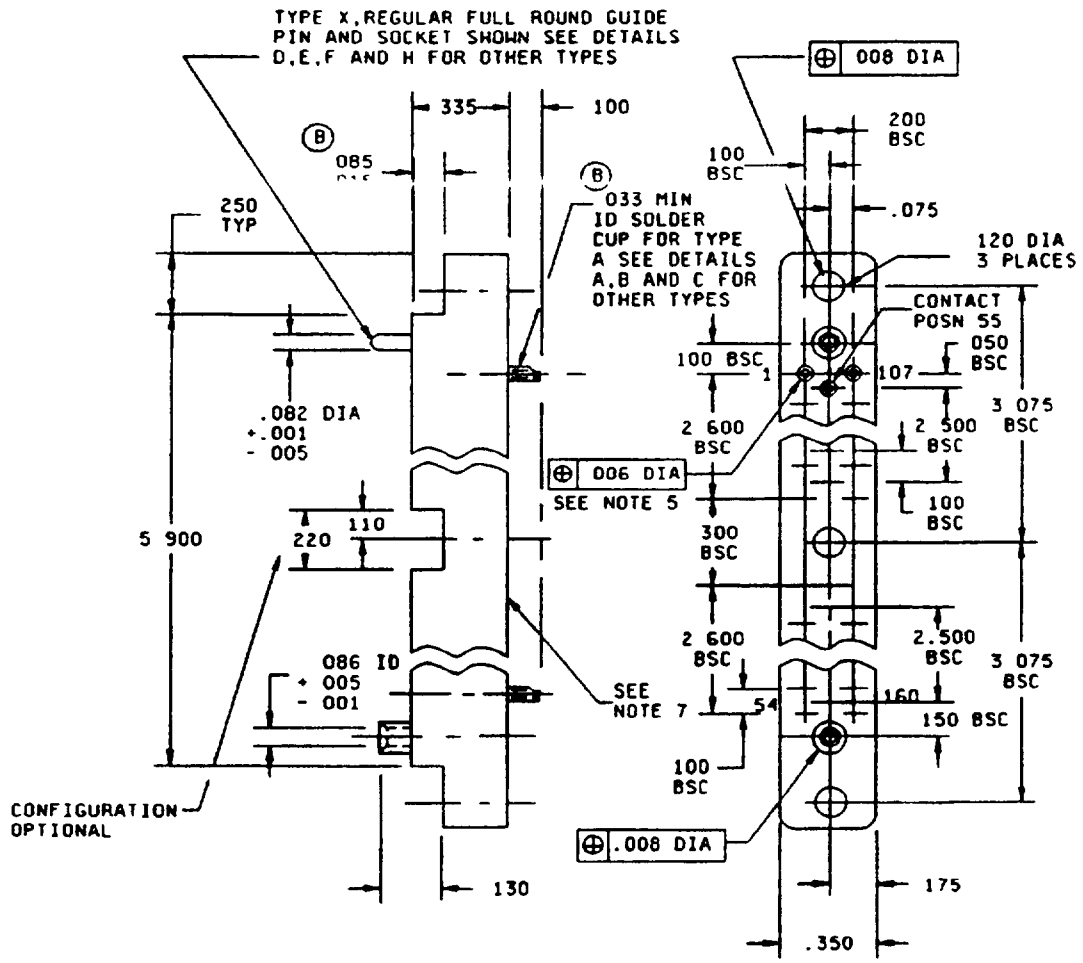
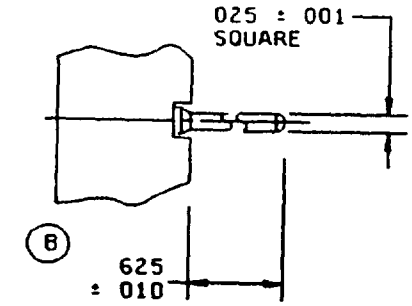


FIGURE 1 Connector, receptacle .100 inch (2 54 mm) spacing.

B denotes changes

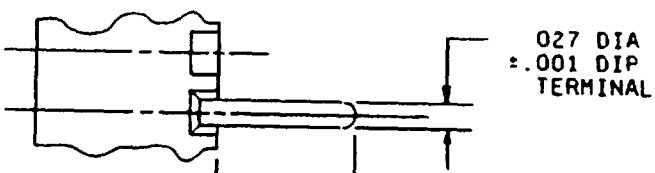
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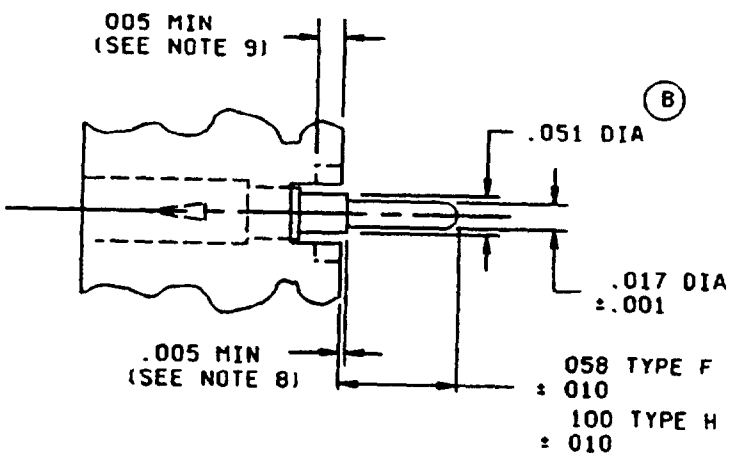
WIRE WRAP TERMINAL (TYPE D)

DETAIL A



172 ± 010 FOR TYPE E DIP TERM
140 ±.010 FOR TYPE C DIP TERM
109 ± 010 FOR TYPE B DIP TERM

DETAIL B

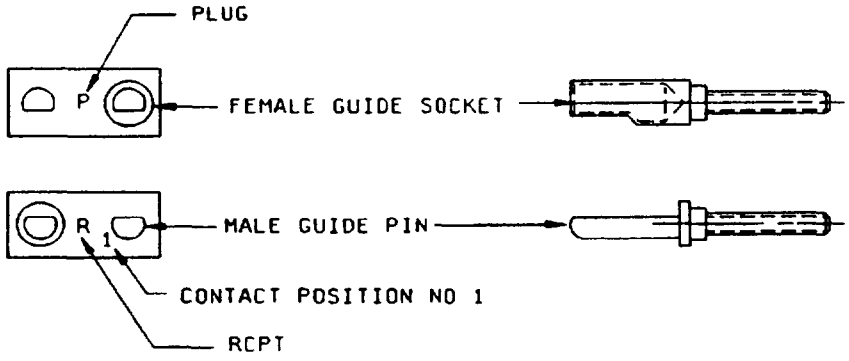


FLEXIBLE CIRCUIT TERMINAL

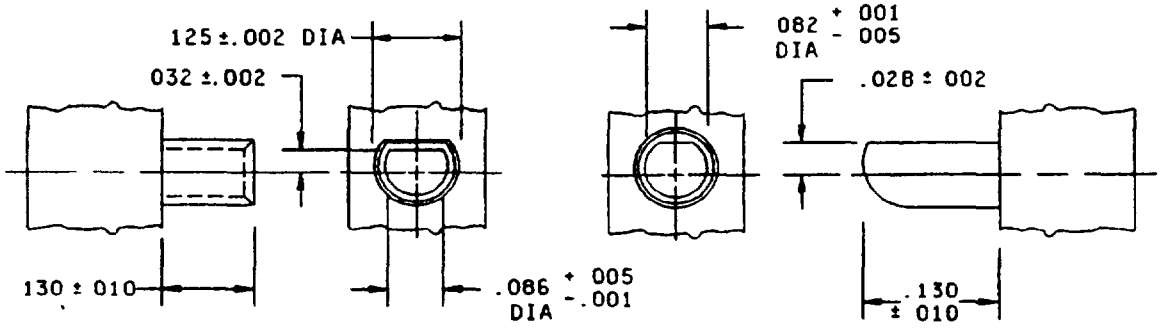
DETAIL C

FIGURE 1 Connector, receptacle 100 (2.54 mm) spacing - Continued

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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS (TYPE Y)



DETAIL D

FIGURE 1 Connector, receptacle 100 inch (2 54 mm) spacing - Continued

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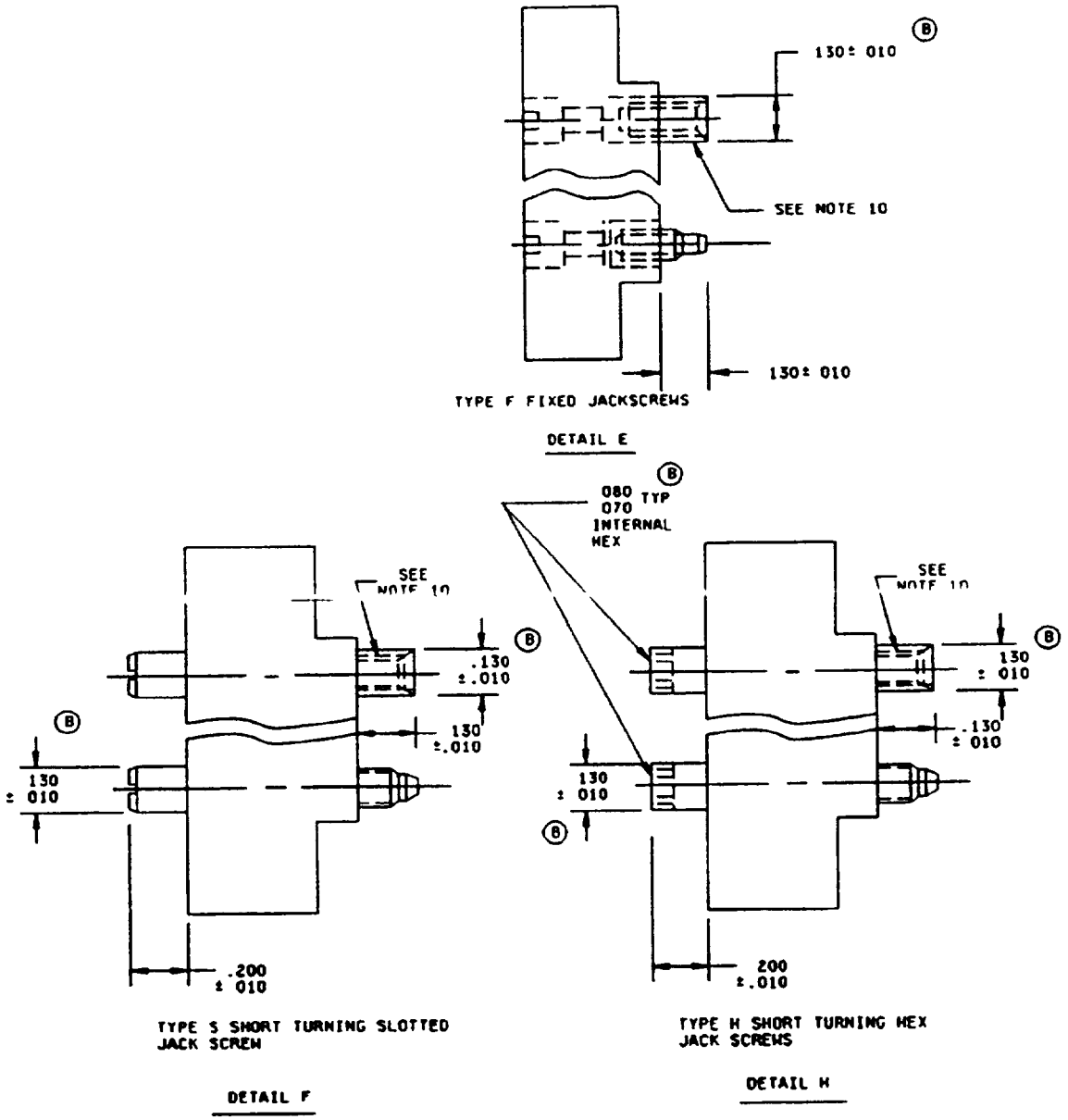


FIGURE 1 Connector, receptacle 100 inch (2 54 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm
001	0 03	094	2 39	2 500	63 50
.002	0 05	100	2 54	2 600	66 04
.005	0.13	109	2.77	3.075	78 10
.006	0.15	110	2 79	5.900	149 86
.008	0.20	120	3 05		
.010	0.25	125	3.18		
.017	0.43	130	3 30		
.025	0.64	.125	3.18		
.027	0.69	130	3 30		
.028	0 71	.140	3.56		
032	0 81	150	3 81		
050	1.27	172	4 37		
051	1 30	175	4 44		
.058	1 47	200	5 08		
.075	1 90	220	5 59		
078	1 98	335	8 51		
.082	2 08	350	8 89		
.086	2 08	.625	15 88		

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm)
4. These connectors mate with connectors specified in MIL-C-55302/139
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .020 (0.51 mm) modular grid.
7. Optional undercut .050 (1.27 mm) maximum, on PCB side for cleaning purposes
8. Conformal coating clearance shall be .005 (.013 mm) minimum (not required for optional design) (B)
9. Optional design conformal coating clearance (B)
10. Hole depth to female guide hardware .282 min (7 16 mm) Full thread depth to female threaded hardware· .240 min (6 1 mm). (B)

FIGURE 1 Connector, plug 100 inch (2 54 mm) spacing - Continued

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REQUIREMENTS

Dimensions and configuration See figure 1

(B) Material Guide pins, type X guide sockets, jackscrews and jack sockets shall be made of corrosion-resisting stainless steel in accordance with ASTM-A582 passivated in accordance with QQ-P-35 or shall be corrosion-resistant copper nickel alloy (61 ±2 percent copper, 25 ±2 percent nickel, 2 5 percent maximum other, and the balance zinc) and shall have a brinell hardness of 145 to 175 Type Y guide sockets shall be beryllium copper in accordance with ASTM B196 and ASTM B197, nickel plated in accordance with QQ-N-290, class 2, 50 to 150 microinches

Plating The contact plating shall be in accordance with MIL-C-55302, or gold in accordance with MIL-G-45204, type II, grade C, class 1 over nickel plating in accordance with QQ-N-290, 30 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, over nickel plating in accordance with QQ-N-290, 30 to 150 microinches

Nonfunctional areas See MIL-C-55302, paragraph 3 3 3 4 3, section C

Contact identification See figure 1

Contact engagement and separation forces (No 22 contacts)

Minimum separation .5 ounce

Maximum engagement. 12.0 ounces

Mating and unmating The maximum mating force, in pounds, shall be the number of contacts multiplied by 0 25 and the withdrawal force, in pounds shall be a minimum of 0 025 times the number of contacts and shall not exceed the measured insert force

Contact rating. 5.0 amperes maximum per contact

Contact resistance: The average contact resistance of all contacts measured shall not exceed 010 ohm, and no individual contact pair shall have a resistance exceeding 020 ohm

Dielectric withstanding voltage

Sea level· 1,000 volts rms

High altitude 300 volts rms

Polarization Type X, regular, guide pin and sockets, full round (see figure 1). Type Y, adjustable D-shaped guide pin and/or socket (see MIL-C-55302/57, detail B) All connectors are supplied in the -1 polarized position. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the original equipment manufacturer by placing the desired polarized position in brackets three spaces after the part number on the purchase order. Example M55302/139-A100Y [-32] Polarization pins and sockets shall be adjusted to any desired position by spanner wrench M55302/57-01. Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three spaces after the part number on the purchase order

Note Hardware with Loctite 83-CV or equivalent applied shall withstand a torque of 7 in-oz applied through the nut

Marking

Connector part number· See example

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Example of part number

M55302/139	A	160	X

Basic number of specification sheet

Type of terminal (see figure 1)

- A = Solder cup
- B = Dip terminal .109 inch length
- C = Dip terminal .140 inch length
- D = Wrappost
- E = Dip terminal .172 inch length
- F = .058 flexible circuit terminal
- H = .100 flexible circuit terminal

Number of contacts (see figure 1)

Type guide pin and/or socket (see figure 1)

- X = Full round
- Y = D-shaped
- F = Fixed jackscrew
- S = .200 slotted turning jackscrew
- H = .200 hex turning jackscrew

CONCLUDING MATERIAL

Review activities:
Air Force - 11, 85, 99

User activity
Air Force - 19

Preparing activity
Air Force - 17

Agent.
DLA - ES

(Project 5935-F407-2)

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